

16BWAY4

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Problem

Let n be a positive integer and let a, b, c, d be integers such that $n|a + b + c + d$ and $n|a^2 + b^2 + c^2 + d^2$. Show that

$$n|a^4 + b^4 + c^4 + d^4 + 4abcd.$$

Solution. Define $P(x) = x^4 - \sigma_1 x^3 + \sigma_2 x^2 - \sigma_3 x + \sigma_4 = (x - a)(x - b)(x - c)(x - d)$.

Observe that $P(a) + P(b) + P(c) + P(d) = 0$. So, $n \mid P(a) + P(b) + P(c) + P(d) =$

$$\left(\sum_{\text{cyc}} a^4 \right) - \sigma_1 \left(\sum_{\text{cyc}} a^3 \right) + \sigma_2 \left(\sum_{\text{cyc}} a^2 \right) - \sigma_3 \left(\sum_{\text{cyc}} a \right) + 4\sigma_4.$$

Since $n \mid \left(\sum_{\text{cyc}} a \right) = \sigma_1$, $n \mid \left(\sum_{\text{cyc}} a^2 \right)$, we have, $n \mid \left(\sum_{\text{cyc}} a^4 \right) + 4abcd$.